DOUGLASCOLLEGE

CSIS 4495 – 002

(Applied Research Project)

Title: **EcoTots**

(MERN Stack Website for the Exchange of Gently Used kid's Clothing)

Progress Report 5

Submitted To:

Prof. Padmapriya Arasanipalai Kandhadai

Submitted By:

Lovish Dhanda (Team Lead)

1. Introduction

Since the last progress report, the primary focus has been on the continued development and integration of the AI chatbot within the EcoTots platform. While the foundation for chatbot functionality has been established, this phase has centered around resolving emerging bugs, refining message flows, and ensuring seamless interaction between the chatbot and the search system. A significant amount of testing has revealed persistent issues affecting the chatbot's consistency, accuracy, and response handling, which are currently being addressed.

This report outlines the ongoing work, the challenges encountered during this implementation phase, and the plans to stabilize and enhance chatbot behavior in upcoming development cycles.

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2. Key Contributions

2.1 AI Chatbot Debugging and Enhancement

- Integrated initial AI chatbot functionality into the frontend and backend.
- Refactored chatbot message flow to improve stability across different use cases.
- Fixed bugs related to broken conversation threads and response mismatches.
- Improved chatbot's ability to handle "Is this still available?" inquiries.
- Conducted tests to ensure chatbot responses align with updated search filters.
- Added temporary error logging mechanisms to track chatbot failures in real-time.
- Enhanced the chatbot training data for better contextual understanding of user queries.

2.2 Search System Refinement (Related to AI Integration)

- Reviewed and modified the search endpoints to ensure compatibility with chatbot queries.
- Fixed minor bugs in filter state management triggered by chatbot-driven searches.
- Improved fallback handling when chatbot fails to return relevant search results.

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3. Challenges Faced and Solutions Implemented

Challenge	**Solution Implemented**
Unexpected chatbot response loops	Introduced conditional flow logic to manage loop exits
Chatbot crashing on invalid input	Implemented input sanitization and fallback messages

Challenge	Solution Implemented
Inconsistent results when chatbot initiated search	Refined backend communication between chatbot and search API
New bugs appearing with each chatbot update	Added detailed logging to identify and isolate issues faster
Difficulty syncing chatbot responses with frontend state	Started modularizing chatbot interactions to simplify debugging

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4. Repo Check-in Summary

Backend Updates:

- chatbot.controller.js: Added logic for handling dynamic product inquiries.
- search.controller.js: Integrated support for chatbot-initiated queries.
- chatbot.middleware.js: Introduced request validation and temporary error logging.

Frontend Updates:

- Chatbot.jsx: Refined UI/UX and fixed issues related to message display and input focus.
- Search.jsx: Updated to support chatbot-triggered queries and highlight responses.

Other:

- Created temporary debug panel (local only) to monitor chatbot errors in dev mode.
- Updated README with known chatbot bugs and developer notes.

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5. Work Log Summary

Date	Hours	Description of Work Done
Mar 31	2.0	Integrated basic chatbot interface and connected to backend
Apr 1	2.5	Fixed initial bugs and added input validation
Apr 2	2.0	Refined chatbot message flow and error handling
Apr 3	2.5	Resolved conversation state mismatch issues

Date	Hours	Description of Work Done
Apr 4	2.0	Enhanced search integration and fallback logic
Apr 5	2.5	Conducted in-depth bug testing and logging improvements
Apr 6	2.0	Continued debugging and prepared error tracker for team use

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6. Future Plans

- Continue resolving existing chatbot bugs and inconsistencies.
- Enhance conversation context retention to improve user experience.
- Implement chatbot fallback options for unsupported queries.
- Introduce smarter intent detection and personalized conversation paths.
- Add real-time analytics dashboard to monitor chatbot performance and error rates.
- Improve integration between chatbot responses and listing interactions (e.g., quick view, save).

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7. Conclusion

This development phase marks significant progress in AI chatbot integration but also reveals the complexity of ensuring a smooth, bug-free experience. While the chatbot is partially functional, continuous bug discovery and iterative fixes are necessary to reach a reliable production-ready state. Focus remains on strengthening the chatbot's capabilities and ensuring it serves as a helpful and intuitive feature within the EcoTots platform.

The ongoing work is pivotal in transforming EcoTots into a smart, user-friendly marketplace that leverages AI to simplify the browsing and communication experience.